



YONG LYHOR

AI Engineer & Data Scientist Student

📍 OCIC Chroy Chongva , Phnom Penh 📞 088 56 888 16 / 096 94 31 319
✉️ yonglyhor1004@gmail.com 🔗 LinkedIn 🐙 GitHub 📠 Telegram 🌐 Portfolio

Professional Profile

I am a fourth-year Data Science student at the Institute of Technology of Cambodia (ITC), with a strong foundation in data science and a passion for real-world applications. Through internships in the education and government sectors, as well as hands-on projects in machine learning, artificial intelligence, data analysis, and full-stack development, I have developed solid technical skills and practical problem-solving abilities. I enjoy working with complex datasets, extracting meaningful insights, and contributing to impactful, data-driven decisions. I am motivated to further develop my skills as a Data Science Intern.

Technical Skills

Data Analysis	Python (Pandas, NumPy, Matplotlib, Seaborn, Keras, SciPy), SQL, Statistical Modeling, Hypothesis Testing
Data Visualization	Power BI (DAX), Matplotlib, Seaborn, Plotly
Machine Learning	Scikit-learn, TensorFlow, Predictive Modeling, Data Preprocessing & Data Cleaning, Feature Engineering and EDA
Business Intelligence	Excel, KPI Development, Dashboard Design, Requirements Analysis
Full Stack Development	Backend: Java Spring Boot (RESTful API), Frontend: HTML, CSS, JavaScript and ReactJS
Databases	MySQL, SQL Server, PostgreSQL, Laragon, Database Design, ETL Processes
Tools	Git, GitHub, Docker, VS Code, Jupyter Notebook, JetBrains, Google Colab
Emerging Technologies	Computer Vision, Large Language Models (LLMs), Prompt Engineering
Collaboration	Teamwork, Communication, Problem-Solving, Time Management

Professional Experience

Part-Time Math Instructor Feb 2025 – Nov 2025
Somnong 12, Phnom Penh

- Taught mathematics to Grade 12 students, delivering structured lessons and exam preparation
- Conducted 2-hour sessions, 5 days per week, simplifying complex concepts and improving problem-solving skills
- Designed exercises, quizzes, and practice tests to reinforce learning outcomes)

Scientific Day 14th 6 Apr 2025
Institute of Technology of Cambodia, Phnom Penh

- Showcased the RagKhmer-Tutor Chatbot project to demonstrate AI-based educational solutions

Backend Developer Trainee (Project Focus) Dec 2025 – Apr 2026
KruIT Solution, ETEC Center, Phnom Penh

- Developed RESTful APIs using Java Spring Boot and ASP.Net for backend services
- Worked part-time (weekends) collaborating with a team to build and maintain system features
- Assisted in database integration, API testing, and debugging to ensure system reliability

Key Projects

Real-Time Sign Language Detection using YOLOv11 Mar 2024 – Jul 2024

- Developed a real-time system to detect 10 sign language gestures using YOLOv11
- Built and labeled a dataset of 900+ images using Roboflow
- Trained and evaluated the model with an 80/20 split, achieving high accuracy 83%

Coffee Aroma — Full-Stack Web Application Jan 2025 – Mar 2025

- Built a full-stack coffee ordering system using Spring Boot and MySQL
- Integrated Telegram Bot API for real-time order notifications
- Developed core features: product catalog, cart, order management, and reviews
- Designed a normalized relational database with multiple tables and constraints
- Created a frontend and admin dashboard using HTML,CSS, and JavaScript

RagKhmer-Tutor Chatbot (LLMs) Jul 2025 – Aug 2025

- Built an AI chatbot that answers questions in Khmer and English using GPT-4o
- Applied RAG to connect the model with a custom Khmer knowledge base
- Improved accuracy answer using multilingual embeddings for Khmer retrieval
- Developed a Streamlit interface with chat history

Cambodia Job Market Dashboard Feb 2026 – Apr 2026

- Scraped 900+ job posts from major Cambodian platforms
- Built interactive Power BI dashboard showing salary comparisons and job trends
- Highlighted most in-demand skills and highest-paying industries
- Used DAX formulas for dynamic salary calculations

Image Caption Generator (AI) Jan 2026 – Apr 2026

- Built an AI model that generates text descriptions for images using Deep Learning (CNN + LSTM)
- Used a pre-trained DenseNet201 model to extract visual features from images (Transfer Learning)
- Generated captions word-by-word using LSTM sequence modeling
- Trained on 8,000+ images with 40,000+ human-written captions
- Developed a Streamlit web app using Streamlit

Education

Bachelor of Engineering in Data Science 2023 – Present
Institute of Technology of Cambodia, Phnom Penh

English Language Program (Part-time) 2024 – Present
Pannāsāstra University of Cambodia, Phnom Penh

Heng Somrin Chork High School Graduated –2022
Grade B (98.710) with commendable performance in the National Exam (BacII)

Additional Qualifications

Languages	Khmer (Native), English (Intermediate)
Business Skills	Requirements Analysis, Stakeholder Communication, Process Optimization
Interests	Predictive Analytics, NLP, LLMs, Cloud Data Solutions

References

PHAUK Sökkhey, Ph.D Deputy Head of Department of Applied Mathematics and Statistics (AMS) & Head of Master Program of Data Science ☎ 012-93-9310 ✉ phauk.sokkhey@itc.edu.kh	KHEAN Vesal Researcher and Lecturer, Department of Applied Mathematics and Statistics ☎ 096-407-5885 ✉ khean.vesal@itc.edu.kh
--	---

COVER LETTER

Yong Lyhor

AI Engineer / Machine Learning / Junior Data Scientist

📍 OCIC Chroy Chongva, Phnom Penh 📞 088 56 888 16 / 096 94 31 319

Dear Hiring Manager,

I am writing to express my interest in the Data Analyst / Junior Data Scientist position at your organization. As a Fourth-year Data Science student at the Institute of Technology of Cambodia (ITC), I have developed a solid foundation in skill data analysis, machine learning, and building practical AI solutions.

Through my academic projects and hands-on experience, I have built real-world applications such as a bilingual AI chatbot using GPT-4o with Retrieval-Augmented Generation (RAG), and a real-time sign language detection system using YOLOv11. These projects strengthened my ability to work with complex datasets, develop predictive models, and deliver solutions that address real-world challenges.

Here's why I believe I would be a strong addition to your team:

- Alignment between your organization's vision and my data science capabilities
- Experience delivering actionable insights with Python, SQL, Power BI, and machine learning
- Passion for using data to solve real-world problems and optimize decision-making

I am eager to apply my technical and analytical skills in a professional environment and contribute to impactful data-driven solutions. I would welcome the opportunity to discuss how I can support your team. Thank you for your time and consideration.

Sincerely,
Yong Lyhor